

Supplement: How Much Can Conjunction Message Histories Predict Beyond Persistence?

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CDM rows $\rightarrow$ cutoff-safe histories $\rightarrow$ event-disjoint split
 $\rightarrow$ candidate models $\rightarrow$ validation selection
 $\rightarrow$ floor hurdle + split conformal $\rightarrow$ frozen test / official test

Fig. 1. From cutoff-safe CDMs to a frozen test. Validation selects the floor hurdle; official-test scores are reports only.

This supplement holds overflow tables and figures for the IEEE conference paper. Confirmatory claims remain in the main text. Code: <https://github.com/ArjunCodess/PRISM>.

I. HYPOTHESIS CHECKLIST

- H1: matched-cohort Δ MAE 4.49 $\rightarrow$ 0.03 — supported, with the 12 h CI covering zero.
- H2: floor-excluded MAE worse than persistence; Wilcoxon $p = 0.91$ — supported.
- H3: ESA L , LOO 66/66, official test — supported under this protocol.
- H4a: log det vs movement $\rho = 0.399$ — univariate yes; partial coefficient reverses after controlling for risk.
- H4b: gap vs floor AUC 0.819 — supported.
- H4c: gap vs movement $\rho = -0.768$ — supported (negative).

II. PROTOCOL

Fig. 1 is the leakage-safe pipeline. All model and policy choices were frozen before official-test evaluation.

III. FEATURE FAMILIES

The shipped feature table has 234 numeric columns: 77 snapshot (CDM fields plus derived geometry and object-type dummies), 68 history summaries, and 88 covariance trends. History stems are risk, miss distance, relative speed, max-risk estimate, and observation counts. Covariance stems are RTN sigmas and log position-covariance determinants. All rows have `time_to_tca` at or above the cutoff. `dilution_gap`

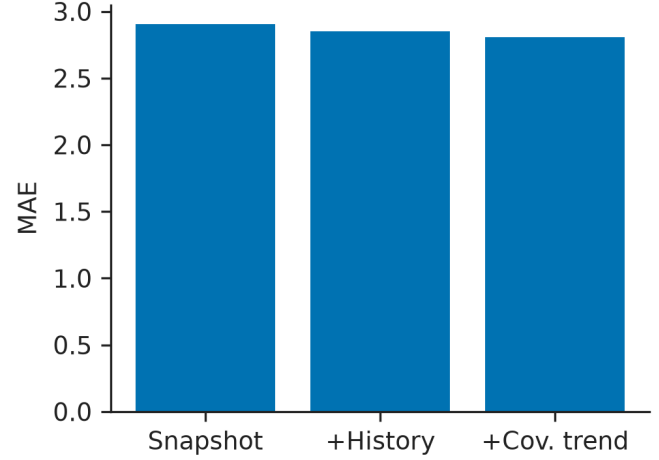


Fig. 2. Unguarded XGBoost MAE: snapshot, plus history, plus covariance trends.

is excluded from the shipped booster. The full name list is `ml/artifacts/feature_schema.json`.

IV. ABLATION

Snapshot MAE 2.904; after history 2.851; after covariance trends 2.808. History adds a small increment relative to the snapshot itself.

V. FLOOR-CLASSIFIER THRESHOLDS

Thresholds 0.05 / 0.10 / 0.15 / 0.20 / 0.30 / 0.50 were scored on the frozen hurdle probabilities without retuning. The validation-selected operating point remains 0.15 (TP 1319, FP 174, FN 33, TN 133). False positives have median later $y = -14.3$ and mean snapshot risk -14.9 .

VI. FIVE-SEED REDRAWS

Seeds 42–46 retrain unguarded and residual XGBoost on grouped event splits. Unguarded MAE advantage 2.34 ± 0.04 ; floor-excluded MAE 7.78 ± 0.21 ; persistence floor-excluded 4.32 ± 0.23 . Seed 42 remains the reported row.

VII. UNMATCHED HORIZON TABLE

The overlapping-set table in earlier drafts mixed cohort change with horizon: advantages 4.534 / 2.272 / 0.524 / 0.060 at 72 / 48 / 24 / 12 h. The matched-cohort table in the main paper is the H1 result.

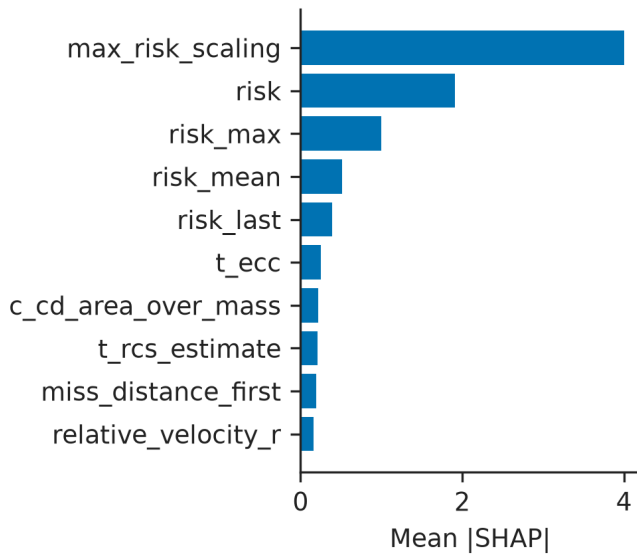


Fig. 3. Mean |SHAP| of the residual regressor on 80 test rows. Risk-trajectory fields dominate; this is not a physical cause.

VIII. GAP QUANTILES

Train-quartile edges of `max_risk_estimate-risk` on the frozen test: Q1 floor rate 0.556 and mean $|y - \text{risk}| = 13.03$; Q3–Q4 floor rates above 0.95 and mean movement below 0.75.

IX. RESIDUAL SHAP

X. REPRODUCIBILITY CHECKLIST

- Code: <https://github.com/ArjunCodess/PRISM>
- Dataset: ESA Kelvins training archive and Zenodo 4463683
- Split seed 42; IDs in `ml/artifacts/split_manifest.json`
- XGBoost: 250 trees, depth 4, learning rate 0.05, subsample 0.85, colsample 0.8, min-child-weight 6, $\ell_1 = 0.1$, $\ell_2 = 2.0$, no early stopping, native NaN
- Floor classifier: 180 trees, depth 3, threshold 0.15, no persist guard
- Figures: `python ml/src/plots.py`
- Official test frozen before look; no retune